Minimal-Shot Autonomous Driving

Geometry-Grounded Runtime, Transfer Diagnostics, WOD-E2E Grounding

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GitHub branch: `CoRL-2027`

This stack is built to handle unfamiliar long-tail traffic structure through explicit geometry, bounded actions, and auditable transfer tests.

Motivation

Autonomy that depends on collecting every rare case will always lag reality.

Minimal-shot autonomy needs a different capability:

Standard AV scaling	Minimal-shot target
train on many mapped cases	reason from first principles
optimize average-case driving	survive long-tail scenarios
hide failures inside aggregate metrics	expose failures by scenario and axis
expensive data dependence	reusable simulator and audit harness

The repo connects runtime, simulator, transfer harness, and real-data grounding in one reproducible pipeline.

System Contributions

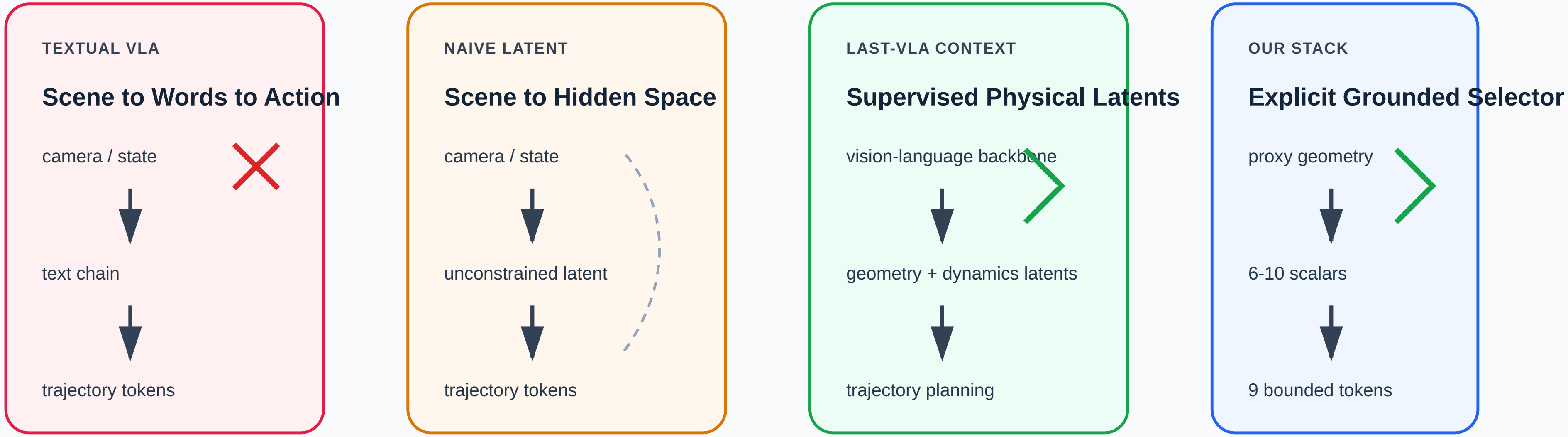
Layer	What we built	Why it matters
runtime	Spotlight Reflex token policy	auditable geometry-first decision path
simulator	custom 2D long-tail environment	controlled causal stress testing
learning probes	BC, RNN, DAgger, scorer variants	isolates what imitation actually learns
transfer harness	AlpaSim proxy-state adapter	exposes external failure modes
grounding track	WOD-E2E selector harness	tests visual/world-model priors on real data

The stack is useful because these layers connect into one evidence chain rather

Why Grounding, Not Language

Driving Reasoning: Text Tokens, Learned Latents, Explicit Grounding

LaST-VLA motivates the shift away from textual reasoning. Our stack implements a smaller explicit version of that idea.



Precise geometry is compressed through language. Latency and hallucination become planning risks.

Fast, but hidden states can drift away from physical geometry and dynamics.

VGGT-like geometry and world-model dynamics are used as teacher priors.

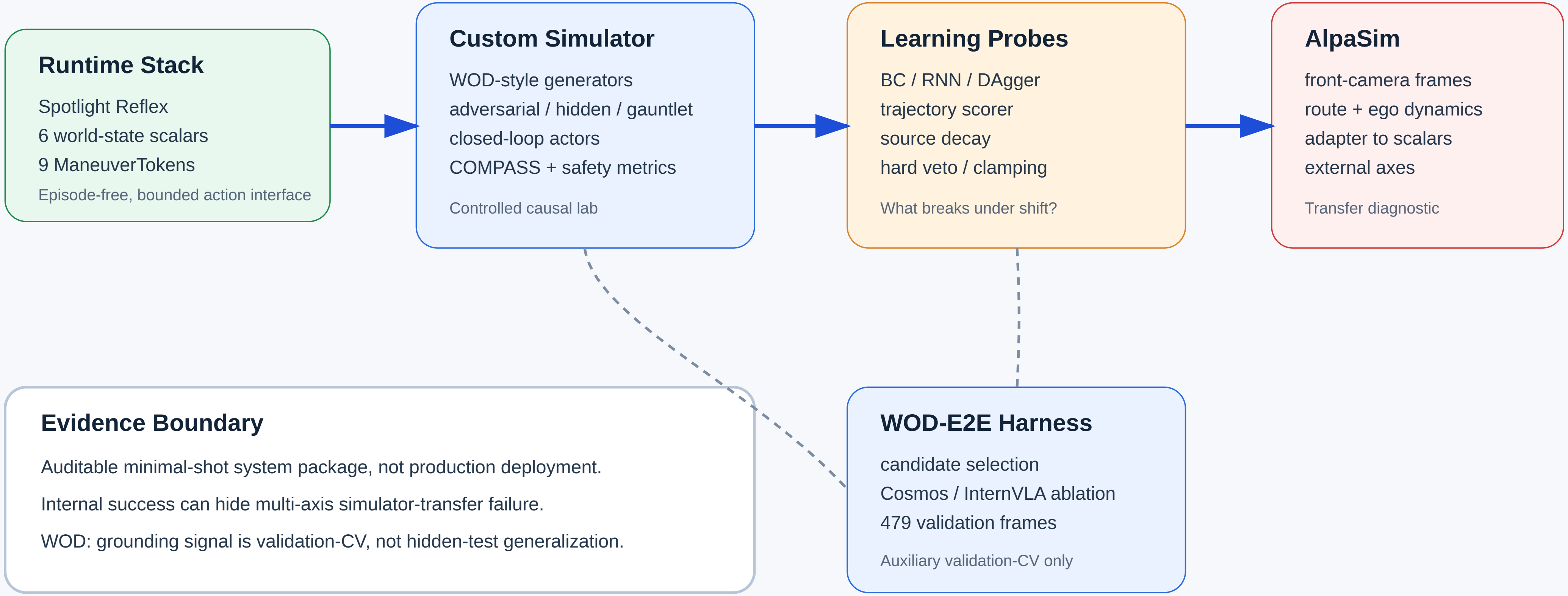
No full VLA training. The physical bottleneck is explicit and auditable.

Adapted claim: we test the same principle at prototype scale: move driving decisions from language-like output toward grounded geometry, dynamics, and bounded action.

Whole System Map

Minimal-Shot AV Stack

Runtime, simulator, transfer harness, and grounding probes share one bounded action interface.



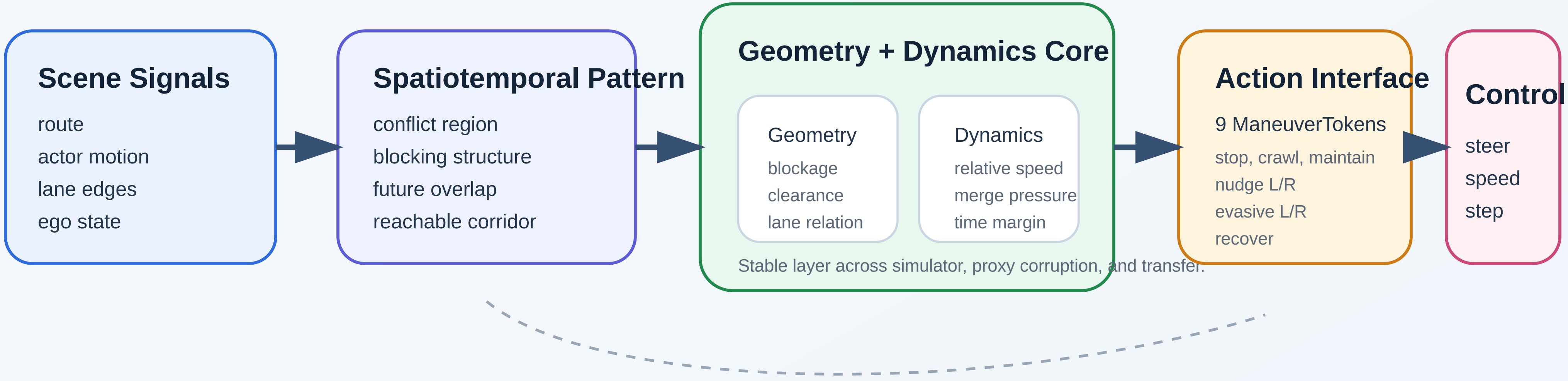
Runtime Architecture

FAST PATH FOR MINIMAL-SHOT DRIVING

SCENE -> SPATIOTEMPORAL PATTERN -> GEOMETRY -> DYNAMICS -> ACTION

Grounded action emerges from spatiotemporal structure.

Compress motion first, resolve geometry and dynamics second, keep the action space bounded at the last step.



Bounded action keeps the last step explicit and auditable.

Explicit Physical Bottleneck

Our “Manifold” Is Explicit: Geometry, Dynamics, Bounded Tokens

We do not train a 124-d latent VLA. We expose the physical bottleneck as small auditable signals and candidate actions.

Scene State

route corridor

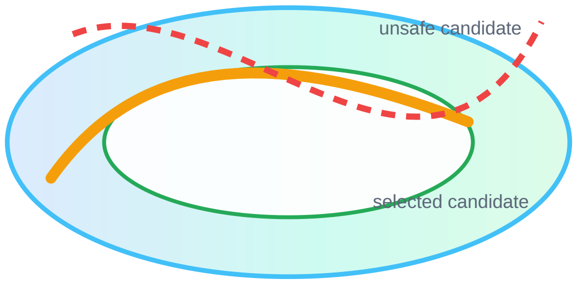
actor positions

relative speed

lane clearance

AlpaSim adapter reconstructs this from camera, route, and ego dynamics.

Explicit Physical Feasibility Region



outer region: route + obstacle geometry

inner region: speed, acceleration, time margin

gold path: selected bounded maneuver

ManeuverToken

stop / crawl

maintain / slow yield

nudge left / right

evasive left / right

lane recover

The final action is a bounded control primitive, not an open language or trajectory string.

Why this matters: the same action manifold can be tested internally, under proxy corruption, and in AlpaSim to see exactly where transfer breaks.

Token Interface

The same action interface is used by the rule policy, learned policies, and AlpaSim transfer experiments.

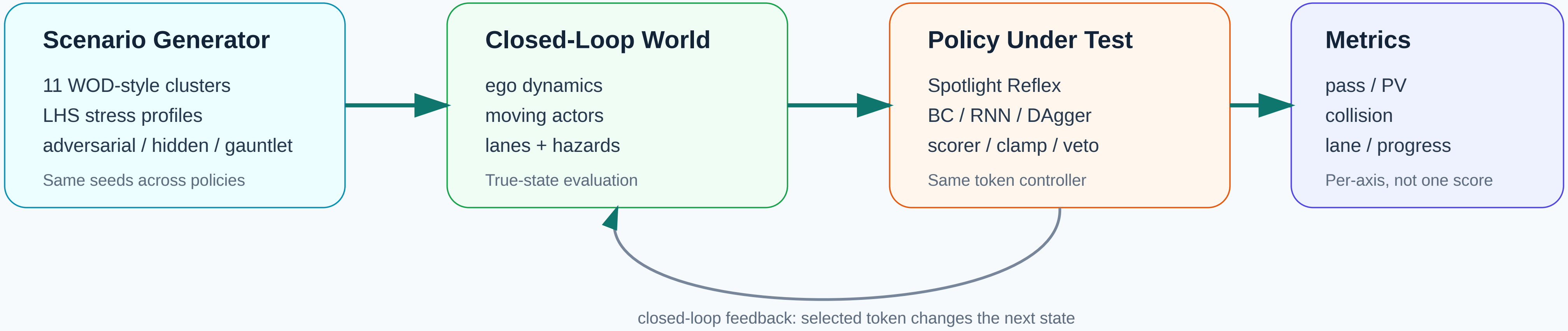
Token family	Examples	Role
longitudinal safety	stop, crawl, slow_yield	reduce collision pressure
nominal progress	maintain	continue along route
mild lateral evasion	nudge_left, nudge_right	create local clearance
emergency evasion	evasive_left, evasive_right	handle close hazards
recovery	lane_recover	return to lane center

This shared token interface makes failures diagnosable: we can separate bad candidate geometry from bad token selection.

Simulator Environment

Custom Simulator: Why It Exists

It is a controlled stress lab where state, proxy corruption, policy, and metrics are independently manipulable.



Proxy Perturbations

heading bias, actor latency
route offset, lane scale
feature noise

Question It Answers

Is AlpaSim failure just an adapter bug?
No: the same tradeoffs appear
under controlled proxy corruption.

Key Output

288 matched proxy cases
1440 agent rollouts
tradeoffs by metric axis

Demo: Minimal-Shot Success And Failure

Baseline failure	Geometry-grounded success
	

Same scenario family: wrong-way actor, low visibility, night.

The baseline extrapolates into the hazard. Spotlight selects an evasive token,

More Long-Tail Scenarios

<table>
<tr><th>Construction zone</th><th>Foreign object debris</th></tr>
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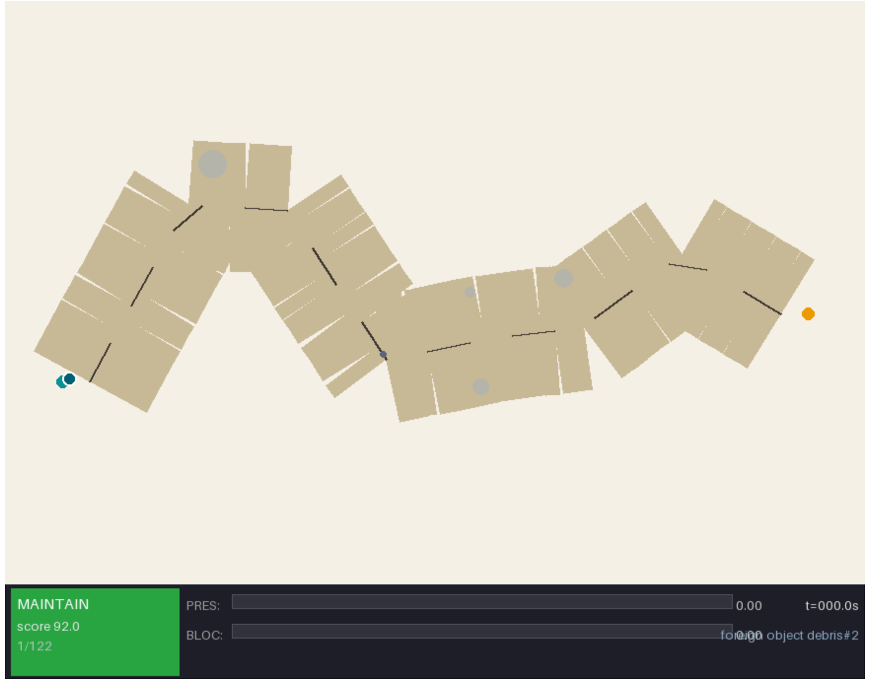
</td><td>
</td></tr>
</table>

The point is controlled variation of geometry, hazards, clearances, and recovery choices.

Wins Snapshot

Win	Result	Why it matters
closed-loop internal safety	`350` rollouts, `0` collisions, `93.1%` overall pass	runtime is not a toy one-off demo
baseline gap on hardest internal suite	`57.6%` vs `2.1%` pass, `7.9%` vs `20.5%` collision	geometry-grounded selection changes behavior materially
real-data grounding signal	`7.845` RFS on WOD-E2E validation-CV	frozen scene priors help when the selector head is right

These are three different wins: internal closed-loop competence, strong baseline separation, and non-trivial grounding signal on real WOD-E2E scenes.

Primary Simulation Result

Closed-loop simulation evidence package:

Suite	Runs	Collisions	Pass
WOD-style, all 11 clusters	110	0	100%
Compositional OOD	60	0	100%
Adversarial	60	0	100%
Hidden holdout	60	0	100%
Gauntlet	60	0	60%
Total	350	0	93.1%

COMPASS: **9.137 / 10** over 700 ranked runs.

Source: `README.md`, `artifacts/compass_evidence_report.json`

Baseline Comparison

Matched gauntlet comparison: same 420 scenarios, same seeds.

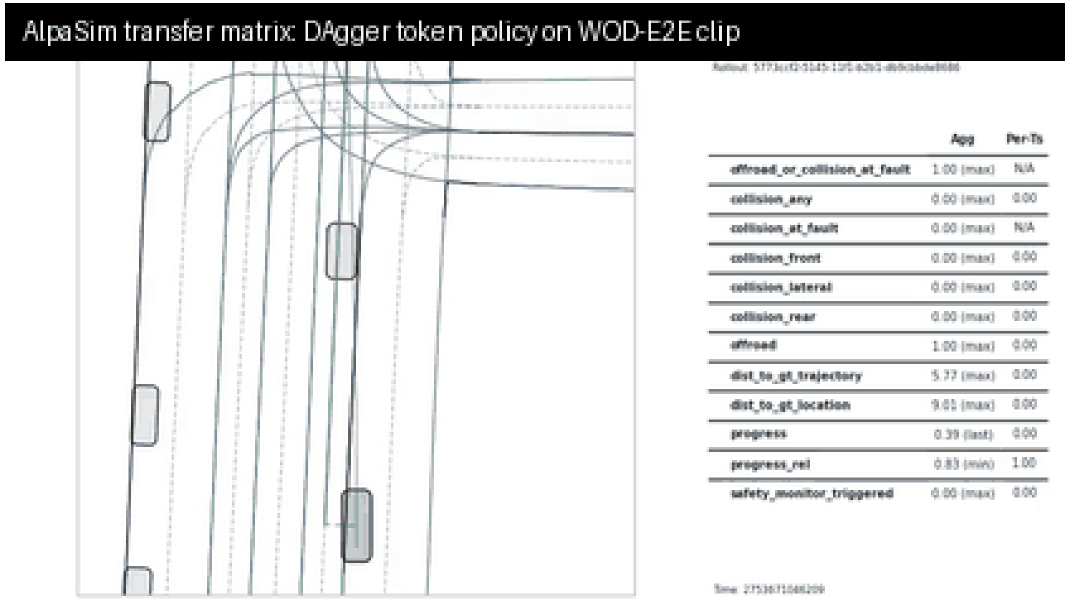
Policy	Pass rate	Collision rate
Baseline, no world-state reasoning	2.1%	20.5%
Spotlight Reflex	**57.6%**	**7.9%**

Explicit geometric reasoning gives a large improvement over a non-reasoning baseline in the randomized challenge environment.

AlpaSim External Transfer

This is a WOD-E2E front-camera AlpaSim rollout with adapter map and metrics.

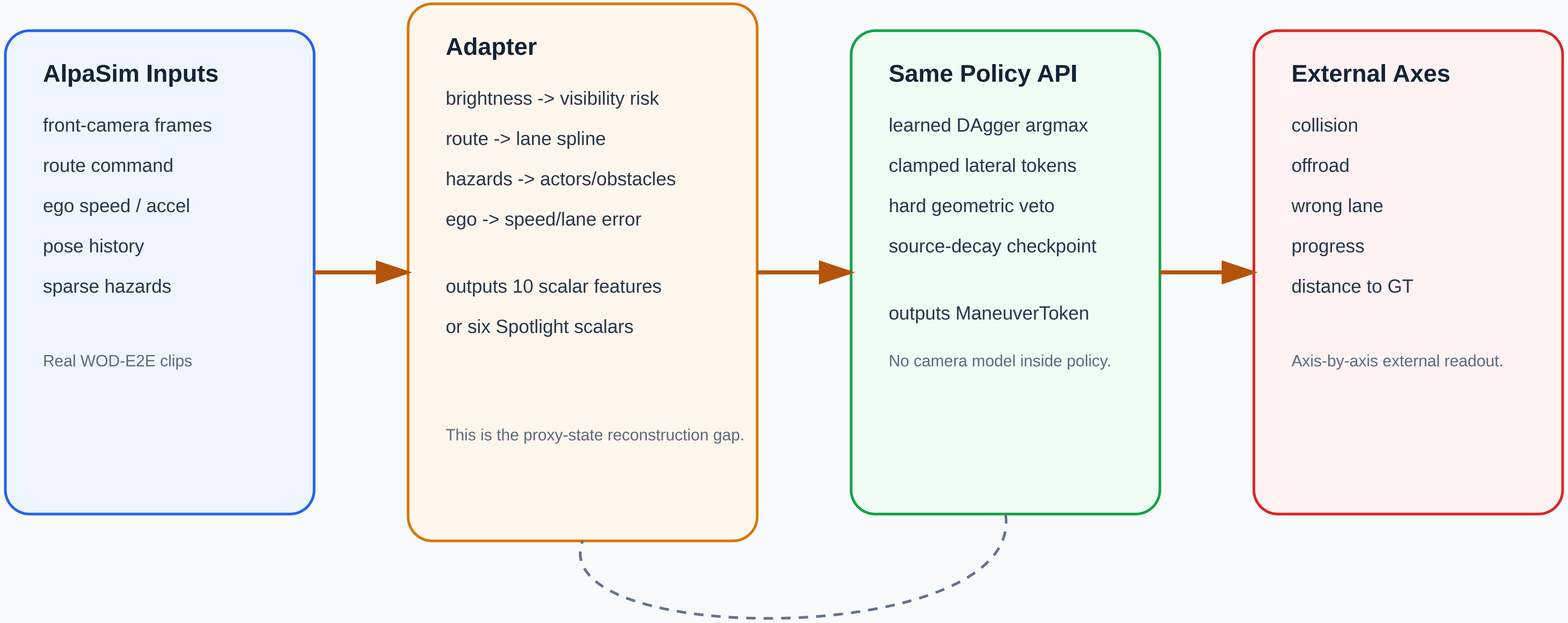
AlpaSim is not used to claim production safety. It is used to test whether the token interface survives a different observation and execution stack.



AlpaSim Adapter

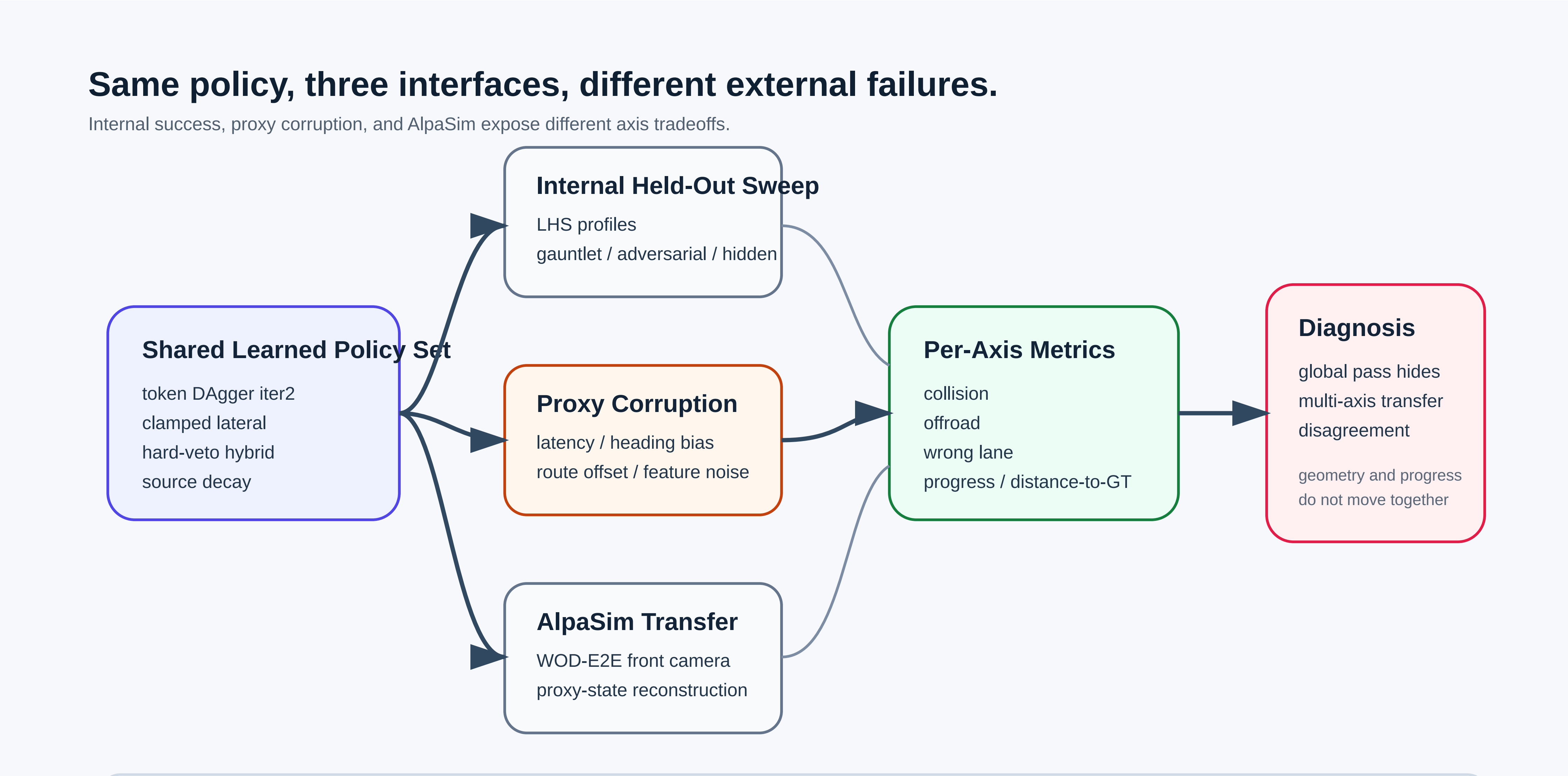
AlpaSim Transfer Stack

The policy is held fixed. Transfer happens through the observation adapter and execution interface.



Selection failures appear only after adapter reconstruction and execution.

Transfer Diagnostic Schema



What Failed Externally

10 matched WOD-E2E clips completed by all four learned variants:

Model	Collision	Offroad	Wrong lane	Progress	Dist. m
Raw DAgger iter2	0.600	0.900	0.700	0.034	61.98
Clamped lateral	0.700	0.500	0.200	0.384	59.85
Hard-veto hybrid	0.800	0.200	0.700	0.837	166.06
Source decay	0.600	0.900	0.400	0.175	62.92

No single learned intervention fixed all axes. This is the main understood failure case: transfer breaks into collision, offroad, lane, and progress axes.

Source: `artifacts/alpasim_matrix10_analysis.md`.

WOD-E2E Auxiliary Track

479 WOD-E2E validation frames, segment-grouped 5-fold CV.

Selector	RFS	Folds	Role
Waymo baseline	7.022	-	reference
Local baseline	7.131	-	reference
Gate-only	7.803	5	selector baseline
RFF direct policy	7.834	5	learned baseline
GPU MLP + Cosmos 64d	**7.845**	5	best validation-CV
Oracle selector	9.264	5	upper bound

This is auxiliary evidence: the right trajectory often exists, but selection still needs stronger grounded scene understanding.

Foundation-Prior Roadmap

Foundation Priors: What We Tested, What Comes Next

LaST-VLA uses teacher priors during VLA training. Our current system uses frozen priors as selector inputs and transfer diagnostics.

BUILT

Explicit Runtime

- 6-10 geometry scalars
- 9 bounded tokens
- trust-region scoring
- scenario generator

Internal: 350 rollouts, 0 collisions, 93.1% pass, COMPASS 9.137.

TESTED

Frozen World Priors

- Cosmos 64d features
- InternVLA 128d probes
- WOD selector CV
- GPU MLP head

Best validation-CV: 7.845 RFS. Signal appears only with the right head.

FOUND GAP

Transfer Mismatch

- proxy-state adapter
- external execution
- axis tradeoffs
- candidate-vs-selector split

AlpaSim exposes collision, offroad, lane, and progress failures separately.

NEXT

Aligned Grounding

- camera hazard encoder
- geometry/dynamics adapters
- axis-aware selector
- 100+ scene external matrix

This is the LaST-VLA-inspired path, adapted to our token-selection stack.

Important boundary: we are not claiming a LaST-VLA implementation. We use its principle to frame our tested Cosmos/InternVLA evidence and the next grounded selector.

Reproducibility

Start here:

```
README.md
presentation.md / presentation-sota.pdf
docs/corl2027/paper.pdf
docs/corl2027/AUDIT.md
./scripts/run_corl2027_audit.sh
```

Primary artifacts:

```
artifacts/minimal_shot_claim_audit.json
artifacts/alpasim_matrix10_analysis.md
artifacts/compass_evidence_report.json
```

Branch:

```
https://github.com/amtellezfernandez/minimal-shot-av/tree/CoRL-2027
```

What This Unlocks

This is now a working testbed for grounded minimal-shot driving.

Built asset	What it enables next
custom simulator with 11 long-tail clusters	controlled stress tests before external runs
explicit geometry-to-token runtime	camera-grounded hazard encoder
BC / DAgger / scorer / veto probes	axis-aware token selector
AlpaSim proxy-state bridge	100+ paired external scenes
WOD-E2E Cosmos / InternVLA probes	preference-aligned visual adapters

Final Takeaway

The core contribution is a minimal-shot AV stack that makes the control path, the stress environment, the transfer interface, and the failure modes all explicit.

The strongest claim is concrete:

the stack works internally, transfers unevenly across external axes, and shows exactly where grounding and proxy-state mismatch start to matter.